

Smart Traffic Violation Analyzer

Project Explanation:

Project Overview:

Smart Traffic Violation Analyzer is a beginner-friendly SQL project that analyzes traffic violation records using basic SQL commands.

The project is designed around the four SQL concepts you have completed on W3Schools:

- SELECT
- SELECT DISTINCT
- WHERE
- ORDER BY

The main purpose is to retrieve traffic violation information, identify unique types of violations, filter records based on specific conditions, and arrange the results in a meaningful order.

Problem Statement

Traffic departments collect information about traffic violations such as vehicle number, vehicle type, violation type, location, fine amount, and violation date.

When there are many records, it becomes difficult to manually identify useful information.

This project uses SQL to answer simple questions such as:

- What traffic violations have been recorded?
- What are the different types of violations?

- Which violations have fines greater than ₹1000?
- Which location has particular violations?
- Which violations have the highest fines?

How the Project Works

Traffic Violation Dataset

- ↓
- Create SQL Table
- ↓
- Insert Data
- ↓
- SELECT Queries
- ↓
- DISTINCT Queries
- ↓
- WHERE Queries
- ↓
- ORDER BY Queries
- ↓
- Analyze Results
- ↓
- Save Output Screenshots
- ↓
- Upload to GitHub

TABLE NAME: Traffic_Violations

CREATE TABLE Traffic_Violations (

```
Violation_ID INT,  
Vehicle_No VARCHAR(20),  
Vehicle_Type VARCHAR(20),  
Violation VARCHAR(50),  
Location VARCHAR(50),  
Fine INT,  
Violation_Date DATE  
);  
  
INSERT INTO Traffic_Violations  
(Violation_ID, Vehicle_No, Vehicle_Type, Violation, Location, Fine, Violation_Date)  
VALUES  
(101, 'TN01AB1234', 'Car', 'Signal Jump', 'Ambattur', 1000, '2026-08-01'),  
(102, 'TN02CD4567', 'Bike', 'No Helmet', 'Avadi', 500, '2026-08-02'),  
(103, 'TN03EF7890', 'Car', 'Overspeeding', 'Anna Nagar', 1500, '2026-08-02'),  
(104, 'TN01GH2345', 'Auto', 'No Parking', 'Ambattur', 750, '2026-08-03'),  
(105, 'TN04IJ6789', 'Bike', 'No Helmet', 'Mogappair', 500, '2026-08-04'),  
(106, 'TN05KL3456', 'Car', 'Overspeeding', 'Avadi', 1500, '2026-08-05'),  
(107, 'TN06MN7890', 'Bike', 'Signal Jump', 'Ambattur', 1000, '2026-08-06'),  
(108, 'TN07OP1234', 'Van', 'No Parking', 'Anna Nagar', 750, '2026-08-07');
```

Traffic_Violations						
Violation_ID	Vehicle_No	Vehicle_Type	Violation	Location	Fine	Violation_Date
101	TN01AB1234	Car	Signal Jump	Ambattur	1000	2026-08-01
102	TN02CD4567	Bike	No Helmet	Avadi	500	2026-08-02
103	TN03EF7890	Car	Overspeeding	Anna Nagar	1500	2026-08-02
104	TN01GH2345	Auto	No Parking	Ambattur	750	2026-08-03
105	TN04IJ6789	Bike	No Helmet	Mogappair	500	2026-08-04
106	TN05KL3456	Car	Overspeeding	Avadi	1500	2026-08-05
107	TN06MN7890	Bike	Signal Jump	Ambattur	1000	2026-08-06
108	TN07OP1234	Van	No Parking	Anna Nagar	750	2026-08-07

SQL Problems for Your Project

1. Display all violation records

SELECT * FROM Traffic_Violations;

Violation_ID	Vehicle_No	Vehicle_Type	Violation	Location	Fine	Violation_Date
101	TN01AB1234	Car	Signal Jump	Ambattur	1000	2026-08-01
102	TN02CD4567	Bike	No Helmet	Avadi	500	2026-08-02
103	TN03EF7890	Car	Overspeeding	Anna Nagar	1500	2026-08-02
104	TN01GH2345	Auto	No Parking	Ambattur	750	2026-08-03
105	TN04IJ6789	Bike	No Helmet	Mogappair	500	2026-08-04
106	TN05KL3456	Car	Overspeeding	Avadi	1500	2026-08-05
107	TN06MN7890	Bike	Signal Jump	Ambattur	1000	2026-08-06
108	TN07OP1234	Van	No Parking	Anna Nagar	750	2026-08-07

2. Display only selected columns

SELECT Vehicle_No, Vehicle_Type, Violation, Fine FROM Traffic_Violations;

Vehicle_No	Vehicle_Type	Violation	Fine
TN01AB1234	Car	Signal Jump	1000
TN02CD4567	Bike	No Helmet	500
TN03EF7890	Car	Overspeeding	1500
TN01GH2345	Auto	No Parking	750
TN04IJ6789	Bike	No Helmet	500
TN05KL3456	Car	Overspeeding	1500
TN06MN7890	Bike	Signal Jump	1000
TN07OP1234	Van	No Parking	750

3. Find unique violation types

SELECT DISTINCT Violation FROM Traffic_Violations;

Violation
Signal Jump
No Helmet
Overspeeding
No Parking

4. Find uniques locations

SELECT DISTINCT Location FROM Traffic_Violations;

Location
Ambattur
Avadi
Anna Nagar
Mogappair

5. Find violations where the fine is greater than Rs.1000

```
SELECT * FROM Traffic_Violations WHERE Fine>1000;
```

Violation_ID	Vehicle_No	Vehicle_Type	Violation	Location	Fine	Violation_Date
103	TN03EF7890	Car	Overspeeding	Anna Nagar	1500	2026-08-02
106	TN05KL3456	Car	Overspeeding	Avadi	1500	2026-08-05

6. Find violations committed in Ambattur

```
SELECT * FROM Traffic_Violations WHERE Locations="Ambattur";
```

Violation_ID	Vehicle_No	Vehicle_Type	Violation	Location	Fine	Violation_Date
101	TN01AB1234	Car	Signal Jump	Ambattur	1000	2026-08-01
104	TN01GH2345	Auto	No Parking	Ambattur	750	2026-08-03
107	TN06MN7890	Bike	Signal Jump	Ambattur	1000	2026-08-06

7. Find bike-related violations

```
SELECT * FROM Traffic_Violations WHERE Vehicle_Type="Bike";
```

Violation_ID	Vehicle_No	Vehicle_Type	Violation	Location	Fine	Violation_Date
102	TN02CD4567	Bike	No Helmet	Avadi	500	2026-08-02
105	TN04IJ6789	Bike	No Helmet	Mogappair	500	2026-08-04
107	TN06MN7890	Bike	Signal Jump	Ambattur	1000	2026-08-06

8. Find violations with a fine of ₹500

```
SELECT *FROM Traffic_Violations WHERE Fine = 500;
```

Violation_ID	Vehicle_No	Vehicle_Type	Violation	Location	Fine	Violation_Date
102	TN02CD4567	Bike	No Helmet	Avadi	500	2026-08-02
105	TN04IJ6789	Bike	No Helmet	Mogappair	500	2026-08-04

9. Sort violations by fine — highest first

SELECT *FROM Traffic_Violations ORDER BY Fine DESC;

Violation_ID	Vehicle_No	Vehicle_Type	Violation	Location	Fine	Violation_Date
103	TN03EF7890	Car	Overspeeding	Anna Nagar	1500	2026-08-02
106	TN05KL3456	Car	Overspeeding	Avadi	1500	2026-08-05
101	TN01AB1234	Car	Signal Jump	Ambattur	1000	2026-08-01
107	TN06MN7890	Bike	Signal Jump	Ambattur	1000	2026-08-06
104	TN01GH2345	Auto	No Parking	Ambattur	750	2026-08-03
108	TN07OP1234	Van	No Parking	Anna Nagar	750	2026-08-07
102	TN02CD4567	Bike	No Helmet	Avadi	500	2026-08-02
105	TN04IJ6789	Bike	No Helmet	Mogappair	500	2026-08-04

10. Sort violations by fine — lowest first

SELECT *FROM Traffic_Violations ORDER BY Fine ASC;

Violation_ID	Vehicle_No	Vehicle_Type	Violation	Location	Fine	Violation_Date
102	TN02CD4567	Bike	No Helmet	Avadi	500	2026-08-02
105	TN04IJ6789	Bike	No Helmet	Mogappair	500	2026-08-04
104	TN01GH2345	Auto	No Parking	Ambattur	750	2026-08-03
108	TN07OP1234	Van	No Parking	Anna Nagar	750	2026-08-07
101	TN01AB1234	Car	Signal Jump	Ambattur	1000	2026-08-01
107	TN06MN7890	Bike	Signal Jump	Ambattur	1000	2026-08-06
103	TN03EF7890	Car	Overspeeding	Anna Nagar	1500	2026-08-02
106	TN05KL3456	Car	Overspeeding	Avadi	1500	2026-08-05

11. Sort vehicles alphabetically

SELECT *FROM Traffic_Violations ORDER BY Vehicle_Type ASC;

Violation_ID	Vehicle_No	Vehicle_Type	Violation	Location	Fine	Violation_Date
104	TN01GH2345	Auto	No Parking	Ambattur	750	2026-08-03
102	TN02CD4567	Bike	No Helmet	Avadi	500	2026-08-02
105	TN04IJ6789	Bike	No Helmet	Mogappair	500	2026-08-04
107	TN06MN7890	Bike	Signal Jump	Ambattur	1000	2026-08-06
101	TN01AB1234	Car	Signal Jump	Ambattur	1000	2026-08-01
103	TN03EF7890	Car	Overspeeding	Anna Nagar	1500	2026-08-02
106	TN05KL3456	Car	Overspeeding	Avadi	1500	2026-08-05
108	TN07OP1234	Van	No Parking	Anna Nagar	750	2026-08-07

12. Find expensive violations in a particular location

SELECT *FROM Traffic_Violations WHERE Location = 'Ambattur' ORDER BY Fine DESC;

Violation_ID	Vehicle_No	Vehicle_Type	Violation	Location	Fine	Violation_Date
101	TN01AB1234	Car	Signal Jump	Ambattur	1000	2026-08-01
107	TN06MN7890	Bike	Signal Jump	Ambattur	1000	2026-08-06
104	TN01GH2345	Auto	No Parking	Ambattur	750	2026-08-03